

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	1 June 2026
Team ID	SWTID-2026-8236
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization :

Develop an accurate, secure, and user-friendly machine learning system that predicts whether a credit card application should be approved or rejected based on applicant information.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & idea prioritization

Project: Credit Card Approval Prediction System

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

- 🕒 10 minutes to prepare
- 🕒 1 hour to collaborate
- 👥 2-8 people recommended

➔ Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

🕒 10 minutes

A Team gathering

Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

Team: Data Analyst, ML Engineer, Frontend Developer, Project Manager

B Set the goal

Think about the problem you'll be focusing on solving in the brainstorming session.

Goal: Build a system that predicts whether a credit card application will be approved or rejected using Machine Learning.

C Learn how to use the facilitation tools

Use the Facilitation Superpowers to run a happy and productive session.

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1 Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

🕒 5 minutes

PROBLEM

How might we predict whether a credit card application will be approved or rejected quickly and accurately using applicant data?



Key rules of brainstorming

To run a smooth and productive session

- 😊 Stay in topic.
- 💡 Encourage wild ideas.
- 🙄 Defer judgment.
- 👂 Listen to others.
- 🔊 Go for volume.
- 👁️ If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping

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Brainstorm

Write down any ideas that come to mind that address your problem statement.

⌚ 10 minutes

TIP You can select a sticky note and hit the pencil icon to quickly jot on your thinking!

Person 1 (Pagolu Srikanth)	Person 2 (Guntupalli Lakshmi)	Person 3 (Peruboina Naga Jyothi)	Person 4 (Mandakini Gurram)
Build ML model to predict credit card approval	Create Flask web application	Perform data pre-processing	Visualize data (EDA)
Use Random Forest algorithm	Design input form for applicant details	Handle missing values & outliers	Compare different ML algorithms
Collect dataset with applicant details	Show approval & rejection probability	Encode categorical variables	Save trained model using joblib
Evaluate model using accuracy, precision, recall	Make UI simple and user-friendly	Feature engineering to improve accuracy	Add validation checks
Deploy model for real-time prediction	Display clear prediction result	Split data into train & test sets	Provide guidance to improve chances

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Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. In the last 10 minutes, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

⌚ 20 minutes

1. Data Collection & Preparation

- Collect credit card approval dataset
- Handle missing values & outliers
- Encode categorical variables
- Feature engineering
- Split data into train & test sets
- Data visualization (EDA)

2. Machine Learning & Model

- Build ML model (Random Forest)
- Compare different ML algorithms
- Tune hyperparameters
- Evaluate model (Accuracy, Precision, Recall, F1 score)
- Save trained model using joblib
- Make predictions on test data

3. Web Application (Flask)

- Create Flask web app
- Design input form for applicant details
- Validate user inputs
- Show approval & rejection probability
- Display clear prediction result
- User-friendly and responsive UI

4. User Insights & Guidance

- Provide factors affecting approval
- Give tips to improve chances
- Show important features
- Explain prediction result clearly
- Help users make informed decisions
- FAQ section for users

5. Deployment & Storage

- Deploy application (PythonAnywhere)
- Store model using joblib
- Ensure app performance
- Backup model and data
- API for prediction
- Secure deployment

6. Testing & Validation

- Test app with sample inputs
- Cross-validate model
- Handle edge cases
- User acceptance testing
- Fix bugs and improve
- Ensure accurate predictions

TIP Add customizable tags to sticky notes to make it easier to find, browse, organize, and categorize everyone's ideas as themes within your mural.

Step-3: Idea Prioritization

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Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

⌚ 20 minutes

The matrix is a 2D grid with 'Importance' on the vertical axis and 'Feasibility' on the horizontal axis. A diagonal line from the top-left to the bottom-right divides the grid into two main regions. The top-left region is labeled 'Low Impact High Feasibility' and 'Easy but less critical - Do if time permits.' The bottom-right region is labeled 'High Impact Low Feasibility' and 'Important but complex - Plan and invest.' The top-right region is labeled 'High Impact High Feasibility' and 'Quick wins - Do these first.' The bottom-left region is labeled 'Low Impact Low Feasibility' and 'Low priority - Consider later or eliminate.'

Importance

If each of these tasks could get done without any difficulty or cost, which would have the most positive impact?

Feasibility

Regardless of their importance, which tasks are more feasible than others? (Cost, time, effort, complexity, etc.)

TIP Participants can use their cursors to point out where they'd most like to focus or copy on. The facilitator can consider this input by using the laser pointer holding the H key on the keyboard.

Tasks Plotted:

- High Impact High Feasibility (Quick wins):** Build ML model (Random Forest) for prediction, User input form for applicant details, Show approval / rejection probability.
- High Impact Low Feasibility (Important but complex):** Real-time credit score integration, Explain prediction using SHAP values.
- Low Impact High Feasibility (Easy but less critical):** Add FAQ section for users, Dark mode interface.
- Low Impact Low Feasibility (Low priority):** Email notifications, Chatbot support.